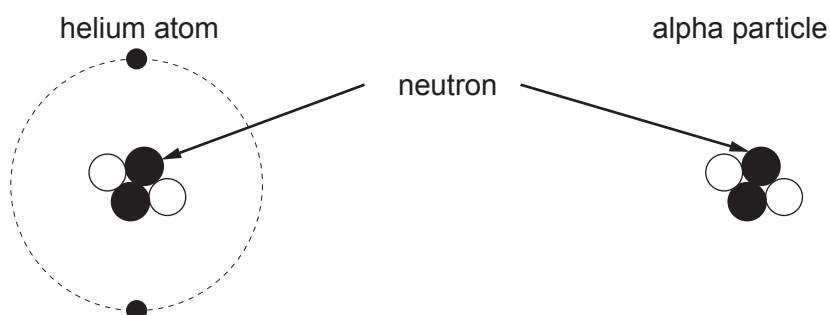


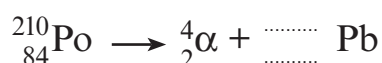
3. A student correctly draws diagrams of a helium atom and an alpha particle.



- (i) One similarity between the structure of the helium atom and the alpha particle has been given in the table below. State **one** other similarity and **one** difference between them. [2]

Similarities	Difference
They both have 2 neutrons.	
.....	
.....	

- (ii) A nucleus of polonium-210 decays by alpha particle emission to form lead (Pb), which is a stable element, according to the following reaction.



Complete the nuclear decay equation above. [2]

- (iii) State why this isotope of polonium is radioactive. [1]

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- (iv) Explain why, in 2006, the Russian Alexander Litvinenko died within weeks of swallowing some polonium-210. [2]

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